Question ID: b75f7812

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Maria plans to rent a boat. The boat rental costs $60 per hour, and she will also have to pay for a water safety course that costs $10. Maria wants to spend no more than $280 for the rental and the course. If the boat rental is available only for a whole number of hours, what is the maximum number of hours for which Maria can rent the boat?

Rationale

The correct answer is 4. The equation $60h + 10 \leq 280$, where h is the number of hours the boat has been rented, can be written to represent the situation. Subtracting 10 from both sides and then dividing by 60 yields $h \leq 4.5$. Since the boat can be rented only for whole numbers of hours, the maximum number of hours for which Maria can rent the boat is 4.

Question ID: 208626df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

$$2\ell + 2w \leq 27$$

A rectangle has length ℓ and width w. The inequality gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to 27. Which statement is the best interpretation of $(\ell, w) = (8, 3)$ in this context?

Answer

A. If the rectangle has length 3 and width 8, its perimeter is less than or equal to 27.

B. If the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.

C. If the rectangle has length 3 and width 8, its perimeter is greater than or equal to 27.

D. If the rectangle has length 8 and width 3, its perimeter is greater than or equal to 27.

Correct Answer: B

Rationale

Choice B is correct. It's given that a rectangle has length ℓ and width w, and the inequality $2\ell + 2w \leq 27$ gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to 27. To determine the best interpretation of $(\ell, w) = (8, 3)$ in this context, the values can be substituted in the given inequality. Substituting 8 for ℓ and 3 for w in this inequality yields $2(8) + 2(3) \leq 27$, which is equivalent to $16 + 6 \leq 27$, or $22 \leq 27$. Since this inequality is true, if the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.

Choice A is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length 8 and width 3, not length 3 and width 8.

Choice C is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length 8 and width 3, not length 3 and width 8.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: f01ef454

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

A geologist estimates that the volume of a slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet. The geologist also estimates that the slab of granite weighs 165 pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the slab of granite?

Answer

A. $165 - 15.7 < x < 165 - 12.7$

B. $165 + 12.7 < x < 165 + 15.7$

C. $165(12.7) < x < 165(15.7)$

D. $\frac{165}{15.7} < x < \frac{165}{12.7}$

Correct Answer: C

Rationale

Choice C is correct. It's given that the estimated volume of the slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet. It's also given that the estimated weight per cubic foot of volume of that slab is 165 pounds. The estimated total weight of the slab of granite, in pounds, can be calculated by multiplying the estimated volume by the estimated weight per cubic foot. Therefore, if the estimated volume of the slab of granite is 12.7 cubic feet, its estimated total weight is $165(12.7)$ pounds, and if the estimated volume of the slab of granite is 15.7 cubic feet, its estimated total weight is $165(15.7)$ pounds. Since the estimated volume of the slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet, the estimated total weight x, in pounds, must be greater than $165(12.7)$ pounds and less than $165(15.7)$ pounds. This situation can be represented by the inequality $165(12.7) < x < 165(15.7)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: e744499e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

An elementary school teacher is ordering x workbooks and y sets of flash cards for a math class. The teacher must order at least 20 items, but the total cost of the order must not be over $80. If the workbooks cost $3 each and the flash cards cost $4 per set, which of the following systems of inequalities models this situation?

Answer

A. $x + y \geq 20$
$3x + 4y \leq 80$

B. $x + y \geq 20$
$3x + 4y \geq 80$

C. $3x + 4y \leq 20$
$x + y \geq 80$

D. $x + y \leq 20$
$3x + 4y \geq 80$

Correct Answer: A

Rationale

Choice A is correct. The total number of workbooks and sets of flash cards ordered is represented by x + y. Since the teacher must order at least 20 items, it must be true that x + y ≥ 20. Each workbook costs $3; therefore, 3x represents the cost, in dollars, of x workbooks. Each set of flashcards costs $4; therefore, 4y represents the cost, in dollars, of y sets of flashcards. It follows that the total cost for x workbooks and y sets of flashcards is 3x + 4y. Since the total cost of the order must not be over $80, it must also be true that 3x + 4y ≤ 80. Of the choices given, these inequalities are shown only in choice A.

Choice B is incorrect. The second inequality says that the total cost must be greater, not less than or equal to $80. Choice C incorrectly limits the cost by the minimum number of items and the number of items with the maximum cost. Choice D is incorrect. The first inequality incorrectly says that at most 20 items must be ordered, and the second inequality says that the total cost of the order must be at least, not at most, $80.

Question ID: 2869fe95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

For a 3-week period in a town in Illinois, the lowest recorded temperature was 31 degrees Fahrenheit $(°\mathrm{F})$ and the highest recorded temperature was $67°\mathrm{F}$. Which inequality is true for any recorded temperature t, in $°\mathrm{F}$, in this town for this 3-week period?

Answer

A. $t \geq 98$

B. $t \geq 67$

C. $31 \leq t \leq 67$

D. $t \leq 31$

Correct Answer: C

Rationale

Choice C is correct. It's given that for a 3-week period in a town in Illinois, the lowest recorded temperature was $31°\mathrm{F}$ and the highest recorded temperature was $67°\mathrm{F}$. It follows that the inequality $31 \leq t \leq 67$ is true for any recorded temperature t, in $°\mathrm{F}$, in this town for this 3-week period.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 84d0d07e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

A clothing store is having a sale on shirts and pants. During the sale, the cost of each shirt is $15 and the cost of each pair of pants is $25. Geoff can spend at most $120 at the store. If Geoff buys s shirts and p pairs of pants, which of the following must be true?

Answer

A. $15s + 25p \leq 120$

B. $15s + 25p \geq 120$

C. $25s + 15p \leq 120$

D. $25s + 15p \geq 120$

Correct Answer: A

Rationale

Choice A is correct. Since the cost of each shirt is $15 and Geoff buys s shirts, the expression $15s$ represents the amount Geoff spends on shirts. Since the cost of each pair of pants is $25 and Geoff buys p pairs of pants, the expression $25p$ represents the amount Geoff spends on pants. Therefore, the sum $15s + 25p$ represents the total amount Geoff spends at the store. Since Geoff can spend at most $120 at the store, the total amount he spends must be less than or equal to 120. Thus, $15s + 25p \leq 120$.

Choice B is incorrect. This represents the situation in which Geoff spends at least, rather than at most, $120 at the store. Choice C is incorrect and may result from reversing the cost of a shirt and that of a pair of paints. Choice D is incorrect and may result from both reversing the cost of a shirt and that of a pair of pants and from representing a situation in which Geoff spends at least, rather than at most, $120 at the store.

Question ID: 89541f9b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

1. $(1, 1)$
2. $(2, 5)$
3. $(3, 2)$

Answer

A. I only

B. II only

C. I and II only

D. I and III only

Correct Answer: C

Rationale

Choice C is correct. Substituting $(1,1)$ into the inequality gives $5(1) - 3(1) < 4$, or $2 < 4$, which is a true statement. Substituting $(2,5)$ into the inequality gives $5(2) - 3(5) < 4$, or $-5 < 4$, which is a true statement. Substituting $(3,2)$ into the inequality gives $5(3) - 3(2) < 4$, or $9 < 4$, which is not a true statement. Therefore, $(1,1)$ and $(2,5)$ are the only ordered pairs shown that satisfy the given inequality.

Choice A is incorrect because the ordered pair $(2,5)$ also satisfies the inequality. Choice B is incorrect because the ordered pair $(1,1)$ also satisfies the inequality. Choice D is incorrect because the ordered pair $(3,2)$ does not satisfy the inequality.

Question ID: 915463e0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Normal body temperature for an adult is between $97.8°F$ and $99°F$, inclusive. If Kevin, an adult male, has a body temperature that is considered to be normal, which of the following could be his body temperature?

Answer

A. $96.7°F$

B. $97.6°F$

C. $97.9°F$

D. $99.7°F$

Correct Answer: C

Rationale

Choice C is correct. Normal body temperature must be greater than or equal to 97.8°F but less than or equal to 99°F. Of the given choices, 97.9°F is the only temperature that fits these restrictions.

Choices A and B are incorrect. These temperatures are less than 97.8°F, so they don't fit the given restrictions. Choice D is incorrect. This temperature is greater than 99°F, so it doesn't fit the given restrictions.

Question ID: df32b09c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G, on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

Answer

A. $90-(85+78+98)\leq 4G$

B. $4G+85+78+98\geq 360$

C. $\frac{(G+85+78+98)}{4}\geq 90$

D. $\frac{(85+78+98)}{4}\geq 90-4G$

Correct Answer: C

Rationale

Choice C is correct. The mean of the four scores (G, 85, 78, and 98) can be expressed as $\frac{G+85+78+98}{4}$. The inequality that expresses the condition that the mean score is at least 90 can therefore be written as $\frac{G+85+78+98}{4}\geq 90$.

Choice A is incorrect. The sum of the scores (G, 85, 78, and 98) isn't divided by 4 to express the mean. Choice B is incorrect and may be the result of an algebraic error when multiplying both sides of the inequality by 4. Choice D is incorrect because it doesn't include G in the mean with the other three scores.

Question ID: 563407e5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

A bakery sells trays of cookies. Each tray contains at least 50 cookies but no more than 60. Which of the following could be the total number of cookies on 4 trays of cookies?

Answer

A. 165

B. 205

C. 245

D. 285

Correct Answer: B

Rationale

Choice B is correct. If each tray contains the least number of cookies possible, 50 cookies, then the least number of cookies possible on 4 trays is 50 × 4 = 200 cookies. If each tray contains the greatest number of cookies possible, 60 cookies, then the greatest number of cookies possible on 4 trays is 60 × 4 = 240 cookies. If the least number of cookies on 4 trays is 200 and the greatest number of cookies is 240, then 205 could be the total number of cookies on these 4 trays of cookies because $200 \leq 205 \leq 240$.

Choices A, C, and D are incorrect. The least number of cookies on 4 trays is 200 cookies, and the greatest number of cookies on 4 trays is 240 cookies. The choices 165, 245, and 285 are each either less than 200 or greater than 240; therefore, they cannot represent the total number of cookies on 4 trays.

Question ID: 2c121b25

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Valentina bought two containers of beads. In the first container 30% of the beads are red, and in the second container 70% of the beads are red. Together, the containers have at least 400 red beads. Which inequality shows this relationship, where x is the total number of beads in the first container and y is the total number of beads in the second container?

Answer

A. $0.3x + 0.7y \geq 400$

B. $0.7x + 0.3y \leq 400$

C. $\frac{x}{3} + \frac{y}{7} \leq 400$

D. $30x + 70y \geq 400$

Correct Answer: A

Rationale

Choice A is correct. It is given that x is the total number of beads in the first container and that 30% of those beads are red; therefore, the expression 0.3x represents the number of red beads in the first container. It is given that y is the total number of beads in the second container and that 70% of those beads are red; therefore, the expression 0.7y represents the number of red beads in the second container. It is also given that, together, the containers have at least 400 red beads, so the inequality that shows this relationship is $0.3x + 0.7y \geq 400$.

Choice B is incorrect because it represents the containers having a total of at most, rather than at least, 400 red beads. Choice C is incorrect and may be the result of misunderstanding how to represent a percentage of beads in each container. Also, the inequality shows the containers having a combined total of at most, rather than at least, 400 red beads. Choice D is incorrect because the percentages were not converted to decimals.